



Ethernet Switching Basics



Foreword

- Data transmission on networks must comply with certain standards. Ethernet protocols define how data frames are transmitted over an Ethernet network. Understanding Ethernet protocols is the basis for fully understanding communication at the data link layer. An Ethernet switch is the main device for implementing data link layer communication. It is essential to understand how an Ethernet switch works.
- This course describes the concepts related to Ethernet protocols, MAC address types, and working process and mechanism of Layer 2 switches.



Objectives

- On completion of this course, you will be able to:
 - ▢ Describe the basic concepts of an Ethernet network.
 - ▢ Distinguish MAC address types.
 - ▢ Get familiar with the working process of a Layer 2 switch.
 - ▢ Get familiar with the structure and generation process of a MAC address table.



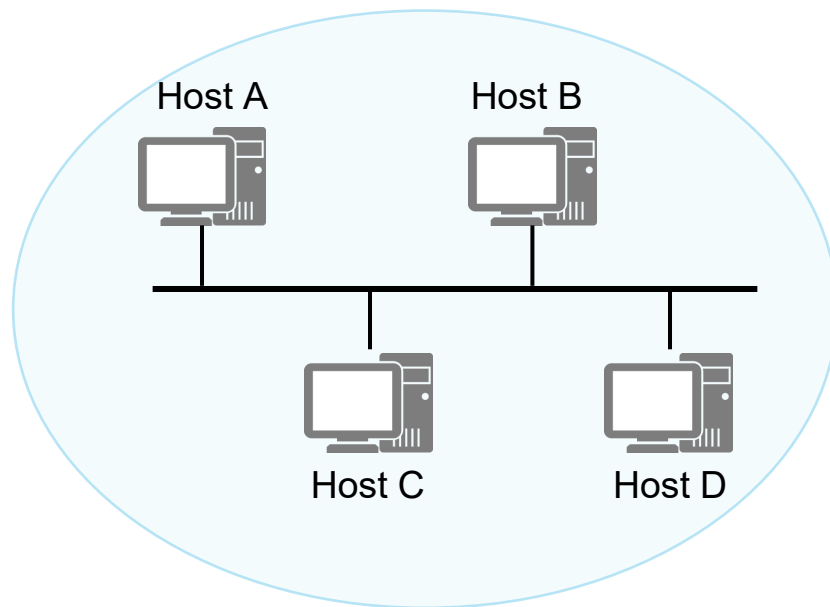
Contents

- 1. Overview of Ethernet Protocols**
2. Overview of Ethernet Frames
3. Overview of Ethernet Switches
4. Process of Data Communication Within a Network Segment

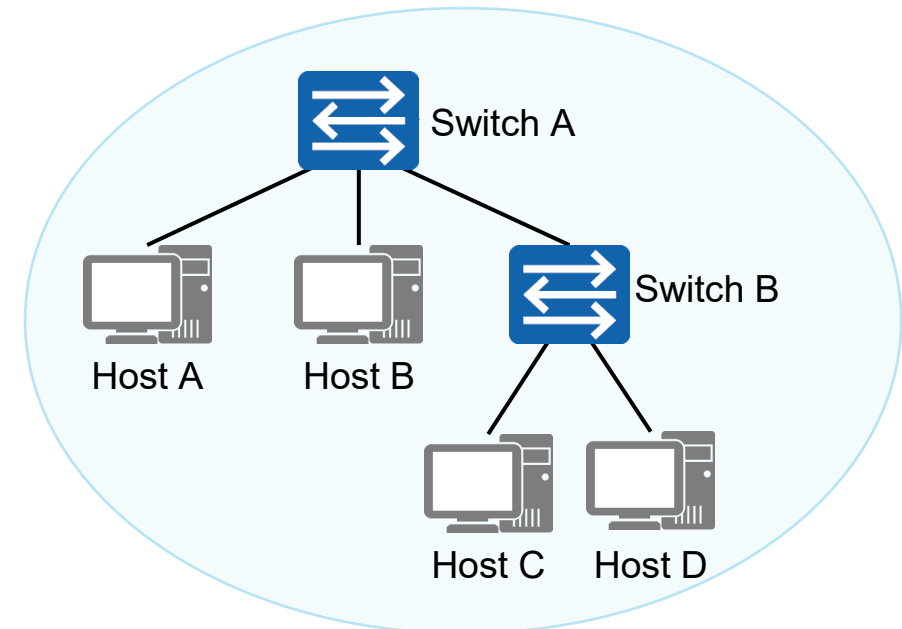
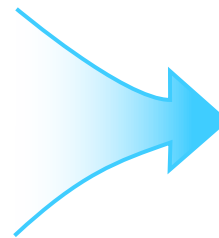


Ethernet Protocols

- Ethernet is the most common communication protocol standard used by existing local area networks (LANs). It defines the cable types and signal processing methods that are used on a LAN.
- An Ethernet network is a broadcast network built based on the carrier sense multiple access/collision detection (CSMA/CD) mechanism.



Early Ethernet

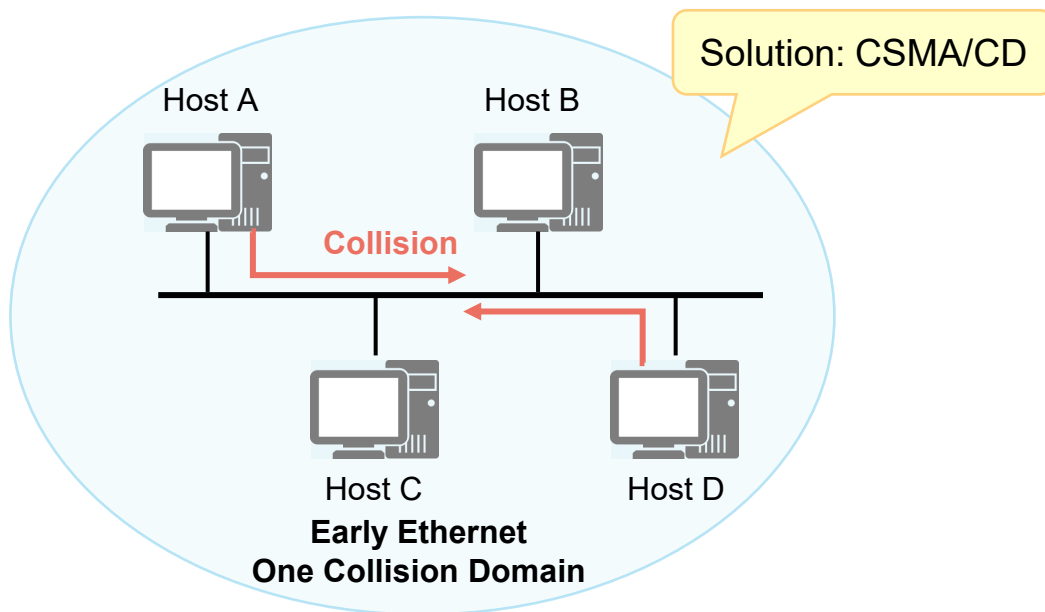


Switch Networking

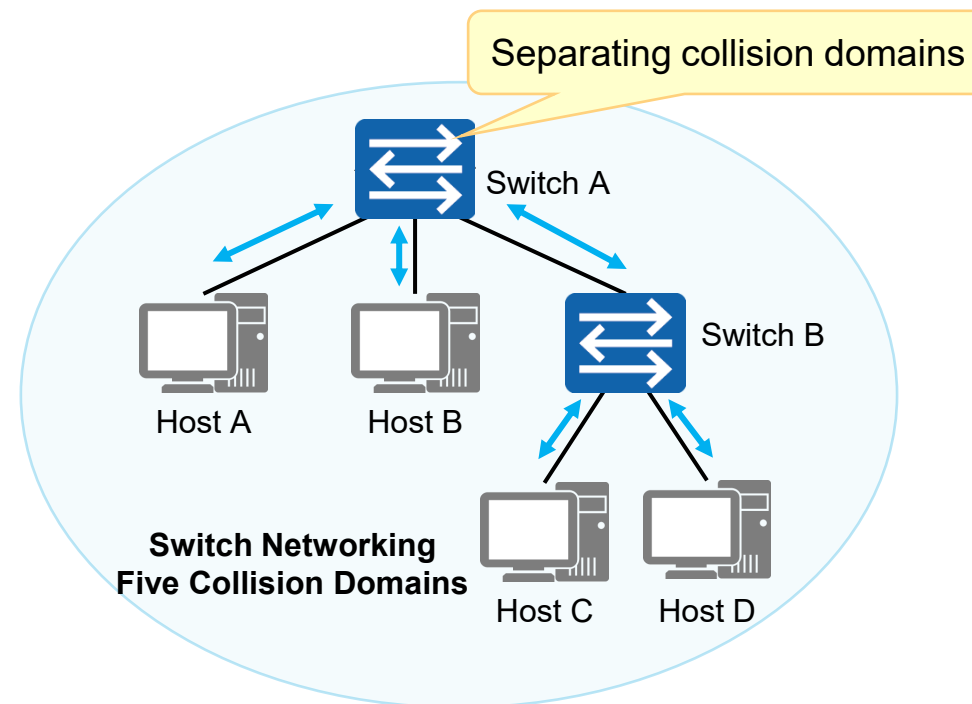


Collision Domain

- A collision domain is a set of nodes connected to the same shared medium. All nodes in a collision domain compete for the same bandwidth. Packets (unicast, multicast, or broadcast) sent by a node can be received by other nodes.



- On a traditional Ethernet network, multiple nodes on the same medium share the link bandwidth and compete for the right to use the link. As a result, collision occurs.
- The probability that collision occurs increases when more nodes are deployed on a shared medium.

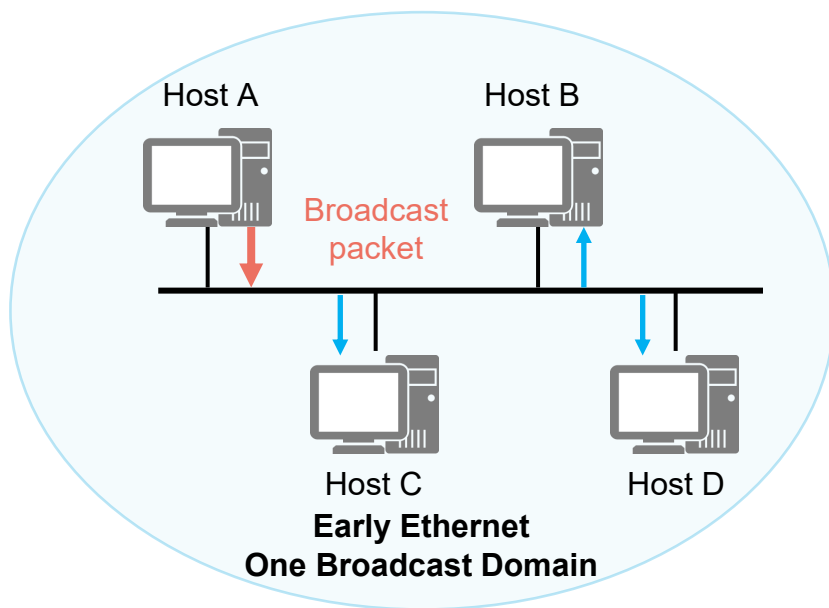


The switch interfaces used to send and receive data are independent of each other and belong to different collision domains. Therefore, collisions do not occur between hosts (or networks) connected through switch interfaces.

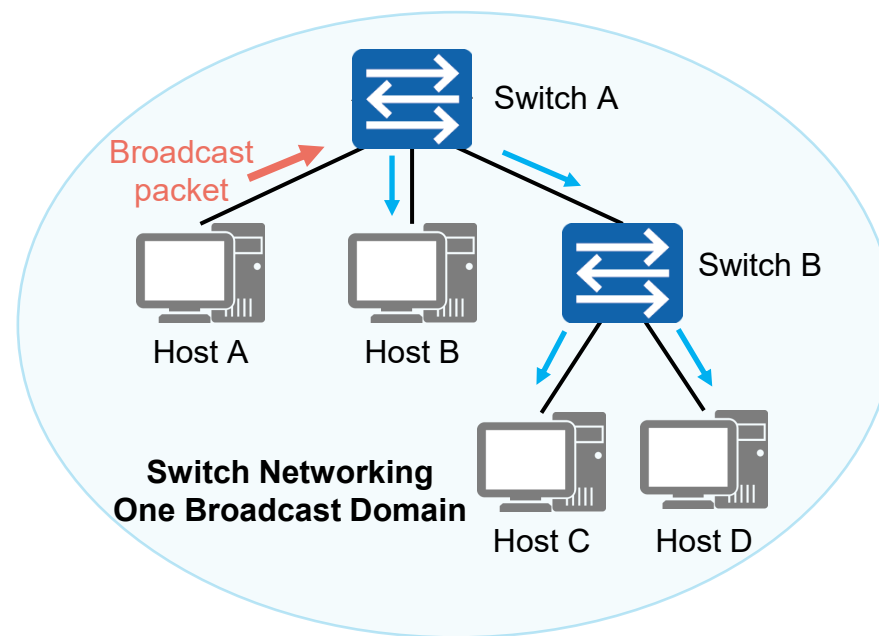


Broadcast Domain

- The entire access scope of broadcast packets is called a Layer 2 broadcast domain, which is also called a broadcast domain. All hosts in the same broadcast domain can receive broadcast packets.



On a traditional Ethernet network, multiple nodes on the same medium share a link. The broadcast packets sent by a device can be received by all the other devices.

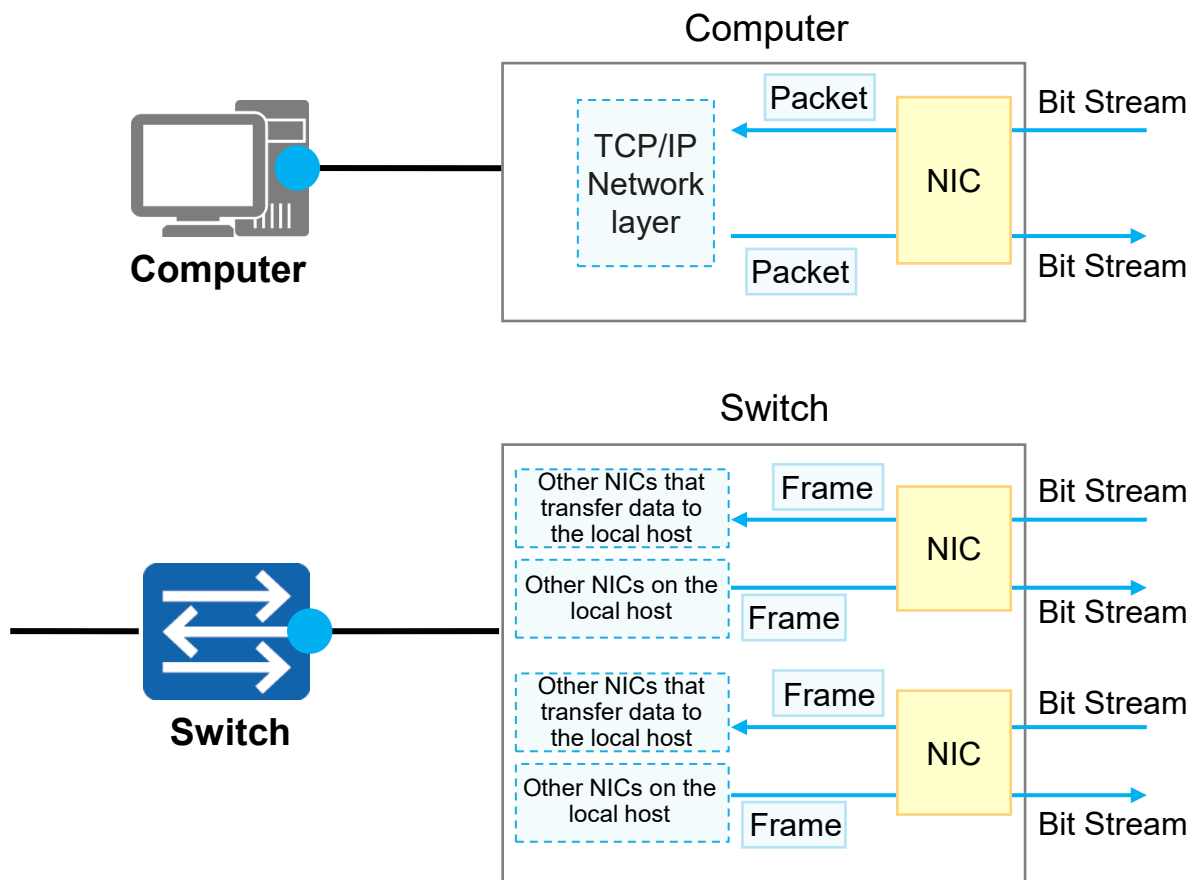


A switch forwards broadcast packets to all interfaces. Therefore, the nodes connected to all interfaces of the switch belong to the same broadcast domain.



Ethernet NIC

- A network interface card (NIC) is a key component that connects a network device (such as a computer, a switch, or a router) to an external network.



- **Network Port**

- A network port is also called a network interface, interface, or port.

- **NIC**

- Each network port corresponds to a NIC.
- A computer or switch forwards data through a NIC.



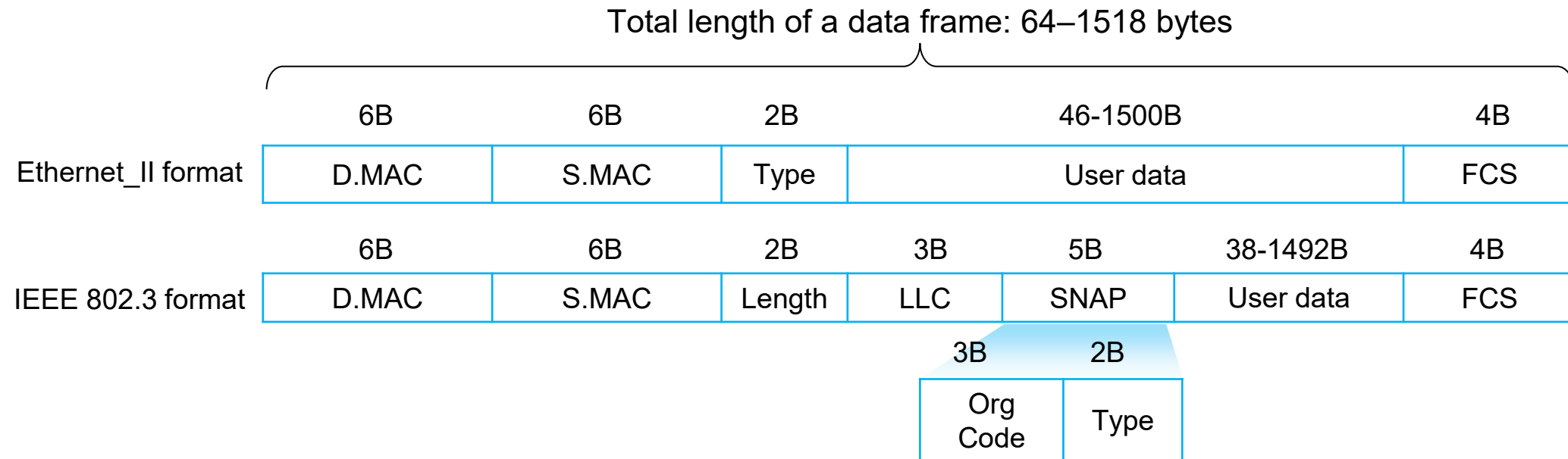
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Ethernet Frame Format

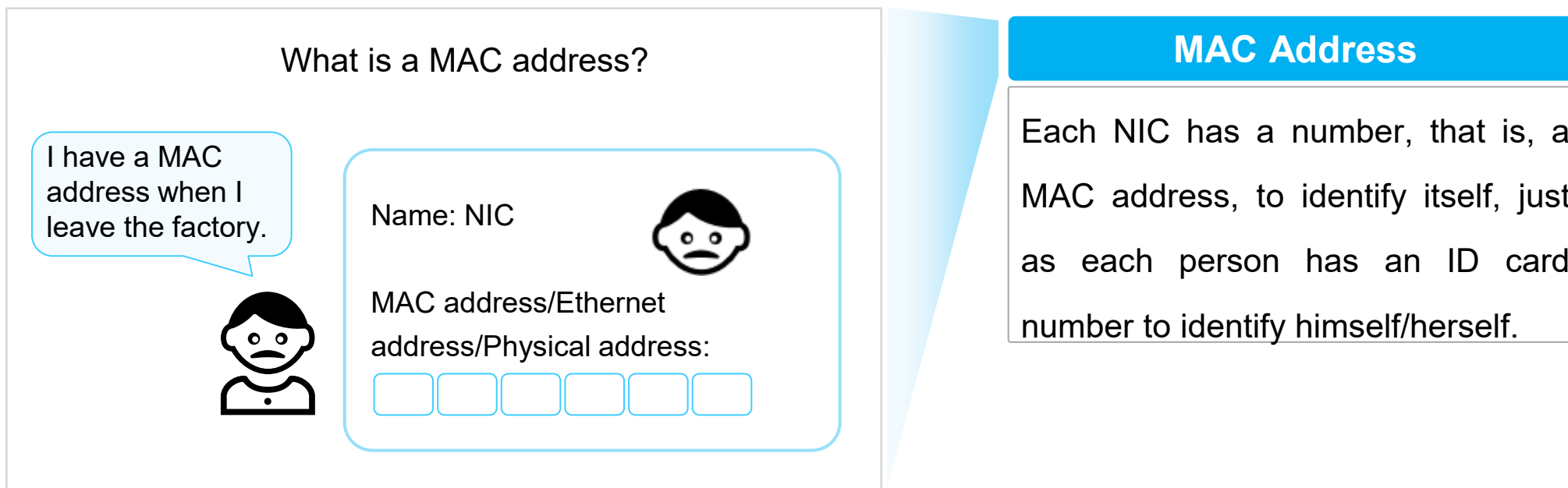
- The frames used by Ethernet technology are referred to as Ethernet frames.
- Ethernet frames are in two formats: Ethernet_II and IEEE 802.3.





What Is a MAC Address?

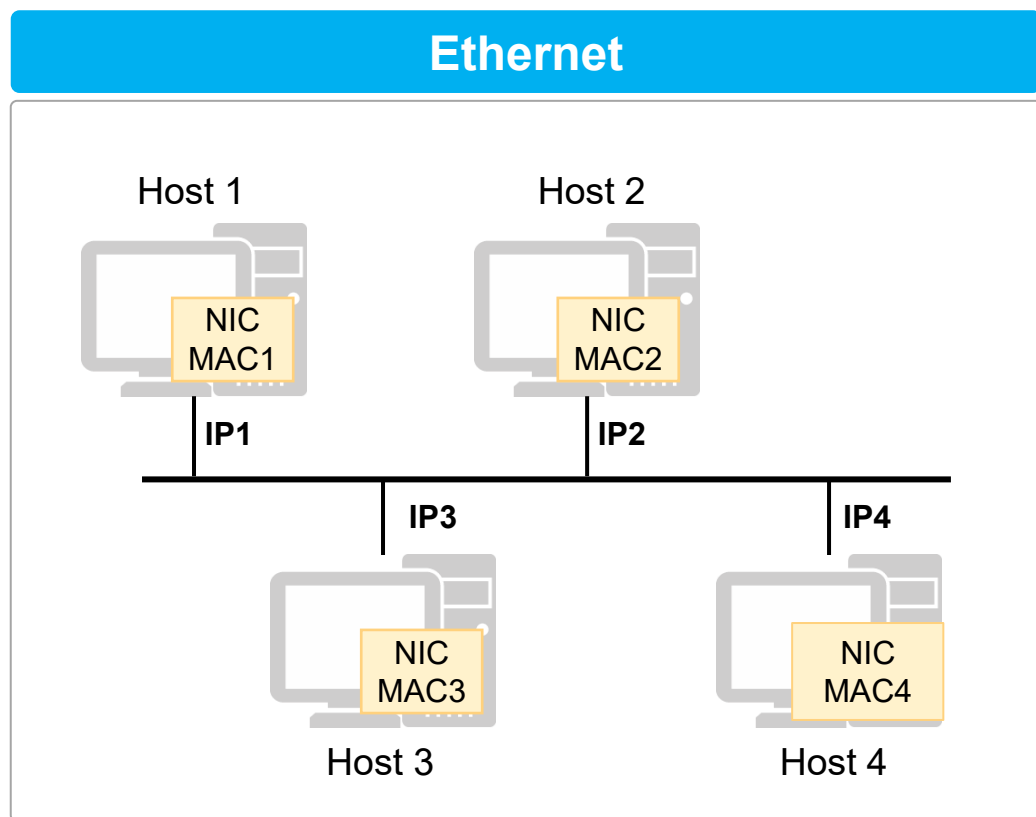
- A media access control (MAC) address uniquely identifies a NIC on a network. Each NIC must have a globally unique MAC address.





IP Address Vs. MAC Address

- Each Ethernet device has a unique MAC address before delivery. When the device accesses the network, it assigns an IP address to each host. Why?



Characteristics of IP addresses:

- IP addresses are unique.
- IP addresses are **changeable**.
- IP addresses are assigned based on **network topology**.

Characteristics of MAC addresses:

- MAC addresses are unique.
- MAC addresses **cannot be changed**.
- MAC addresses are assigned based on the **manufacturer**.

Can a network device have either a MAC address or an IP address?





MAC Address Presentation

- A MAC address is 48 bits (6 bytes) in length.
- As typically represented, MAC addresses are recognizable as six groups of two hexadecimal digits, separated by hyphens, colons, or without a separator.

For example, 00-1E-10-DD-DD-02 or 001E-10DD-DD02

Hexadecimal	00	1E	10	DD	DD	02	6-byte
Binary	0000 0000	0001 1110	0001 0000	1101 1101	1101 1101	0000 0010	48-bit

Conversion
between
hexadecimal and
binary digits

Power	2 ³	2 ²	2 ¹	2 ⁰	2 ³	2 ²	2 ¹	2 ⁰
	8	4	2	1	8	4	2	1
Bit	0	0	0	1	1	1	1	0
	= 1				= 8+4+2=14=E			

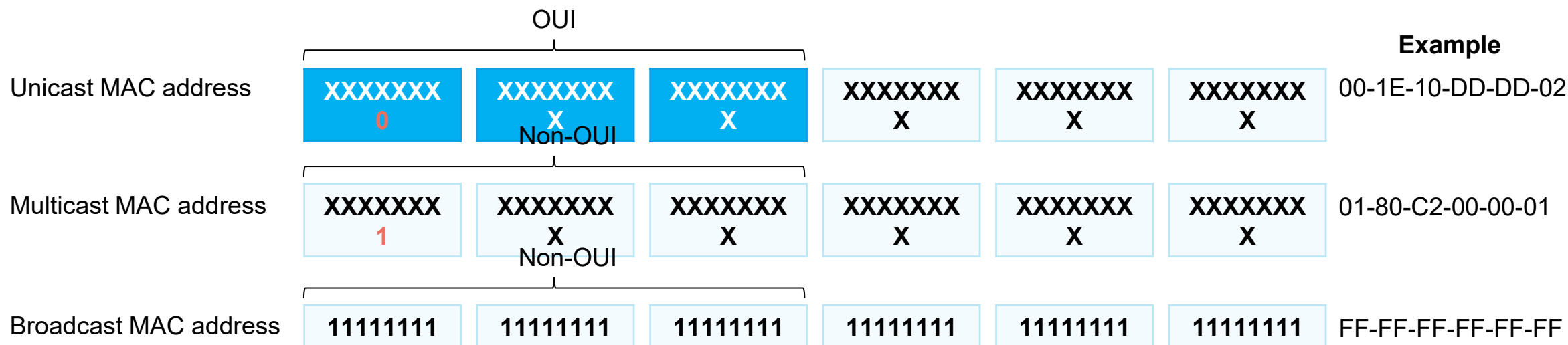


MAC Address Composition and Classification

- Organizationally unique identifier (OUI): a 24-bit (3-byte) number. It is a globally unique identifier assigned by the IEEE.
- Company ID (CID): a 24-bit (3-byte) number. It is assigned by a manufacturer.



- MAC address classification:





Unicast Ethernet Frame

- A unicast Ethernet frame is also called a unicast frame.
- The destination MAC address of a unicast frame is a unicast MAC address.

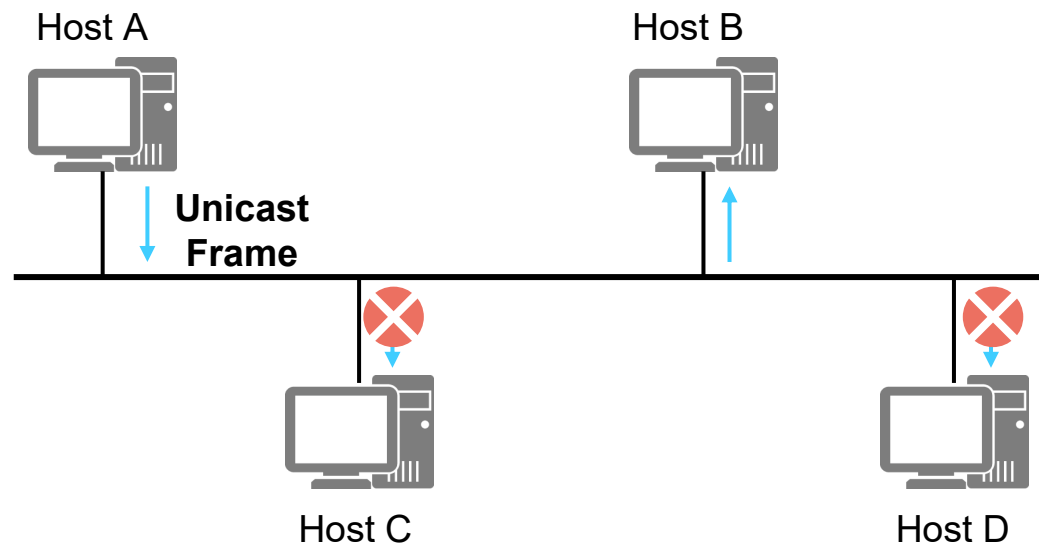


D.MAC:
00-1E-10-DD-DD-02

S.MAC

DATA

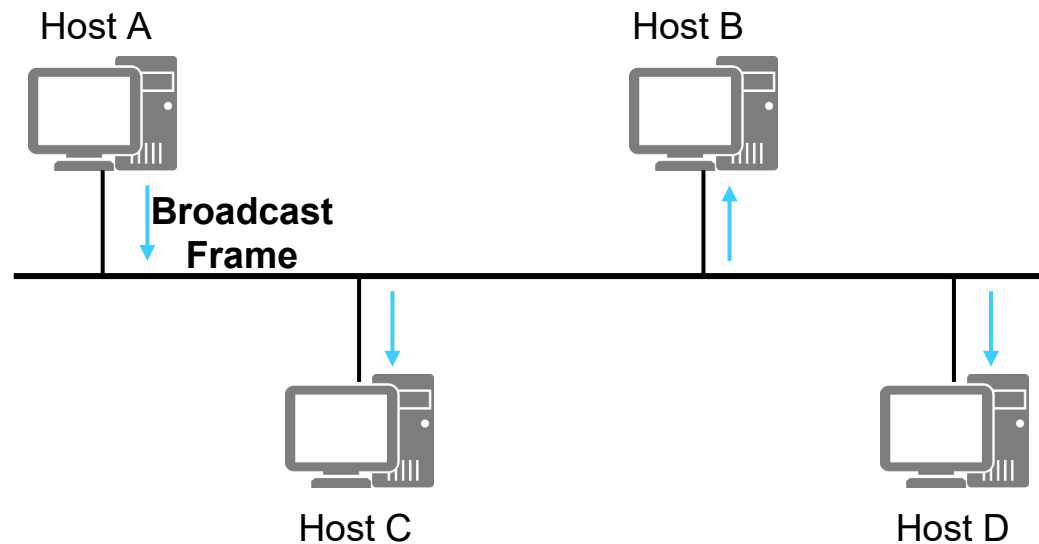
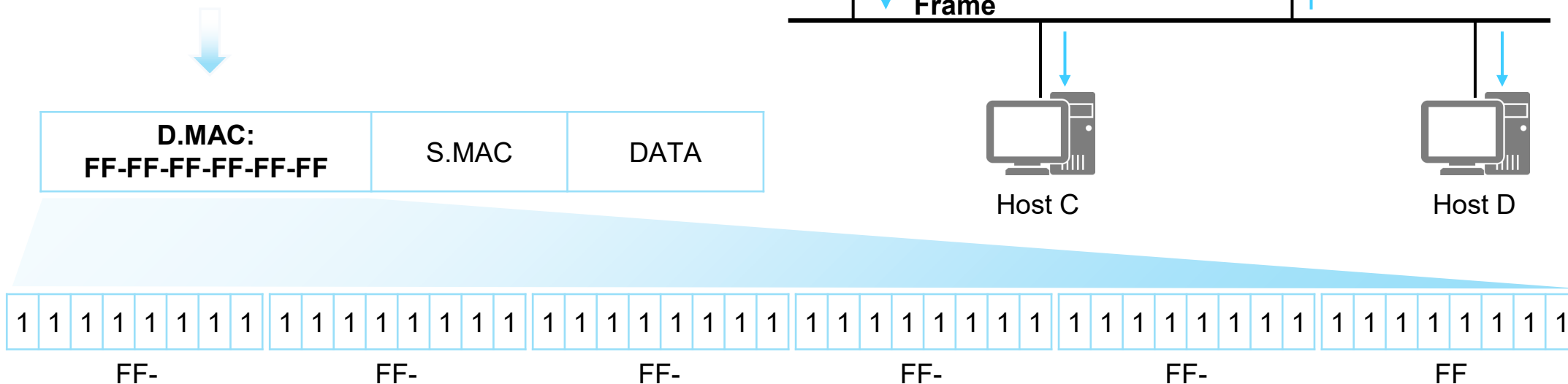
0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	1	0	0	0	0	1	1	0	1	1	1	0	1	1	1	0	1	0	0	0	0	0	1	0							
00-								1E-								10-								DD-								DD-								02							





Broadcast Ethernet Frame

- A broadcast Ethernet frame is also called a broadcast frame.
- The destination MAC address of a broadcast frame is a broadcast MAC address.





Multicast Ethernet Frame

- A multicast Ethernet frame is also called a multicast frame.
- The destination MAC address of a multicast frame is a unicast MAC address.



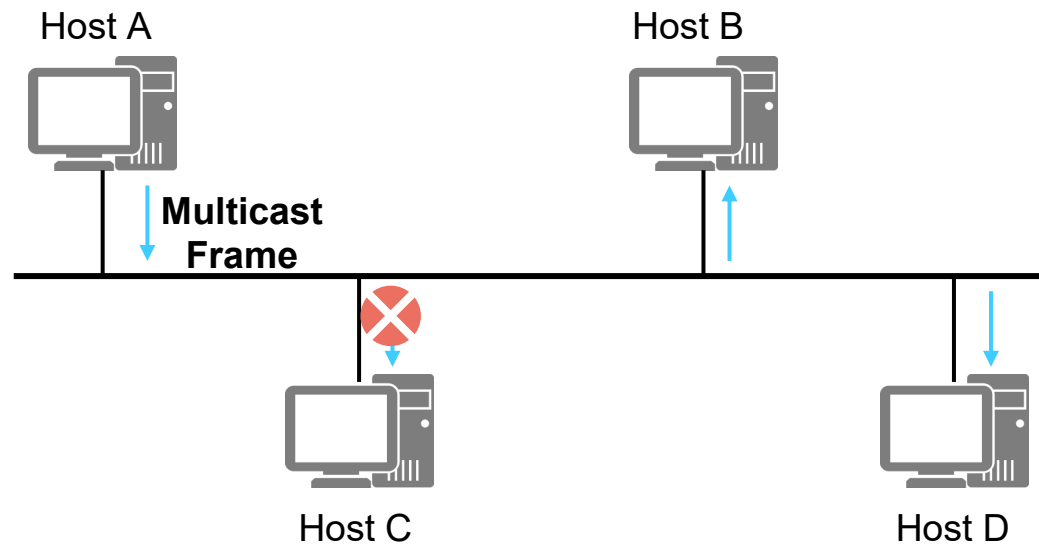
D.MAC:
01-80-C2-00-00-01

S.MAC

DATA

0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 1 1 0 0 0 0 1 0 1

01- 80- C2- 00- 00- 01



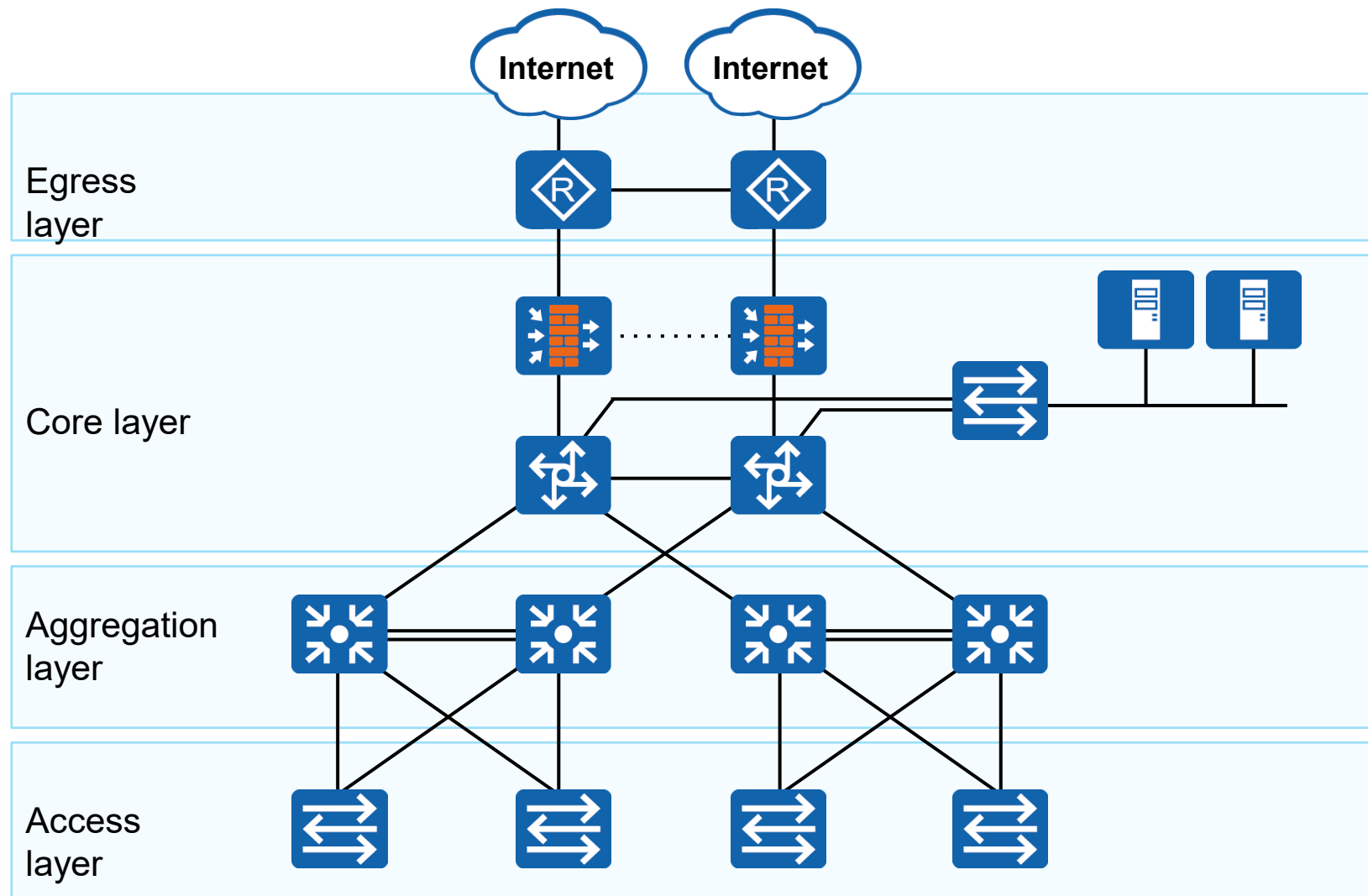


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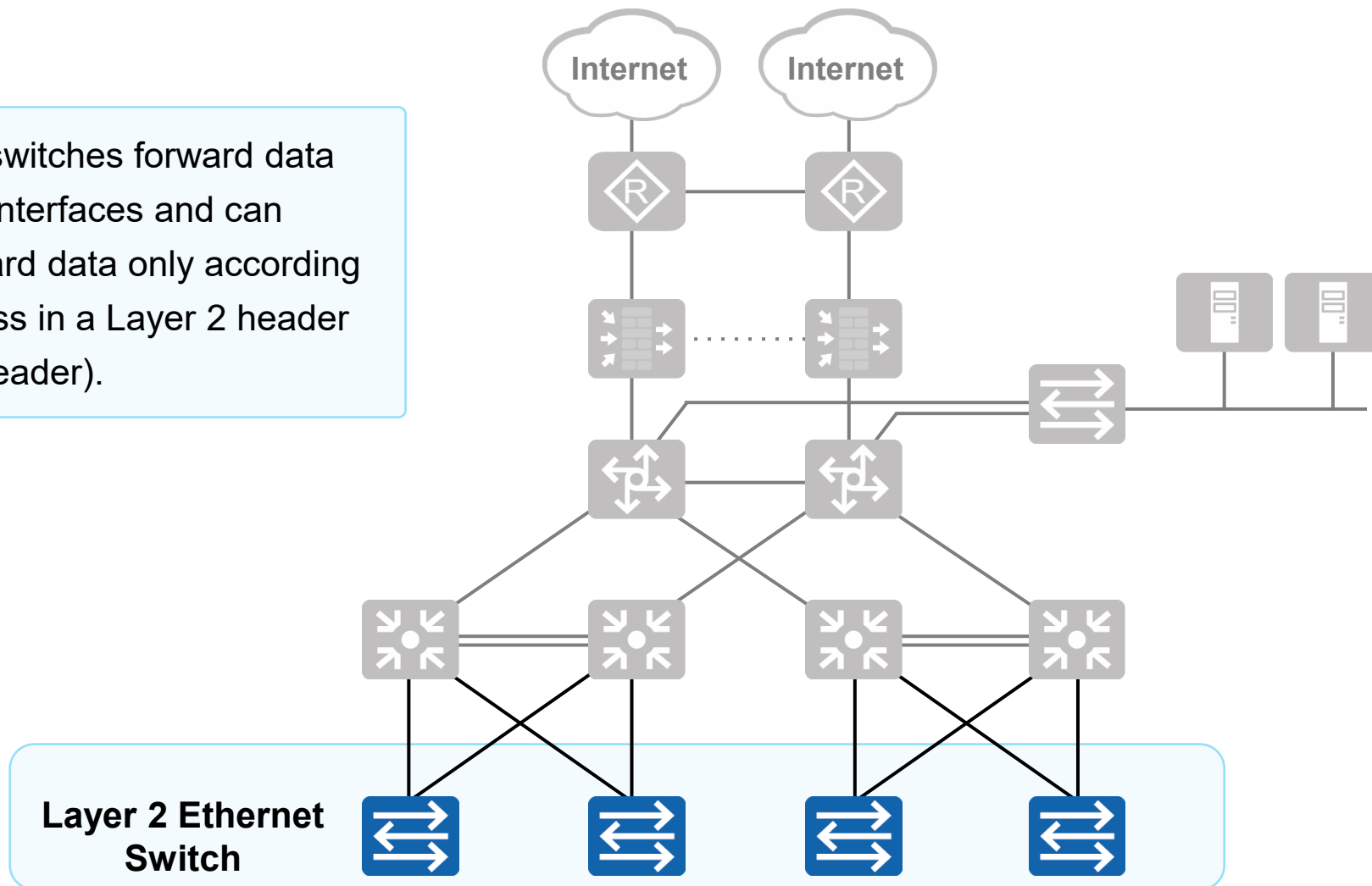
Typical Architecture of a Campus Network





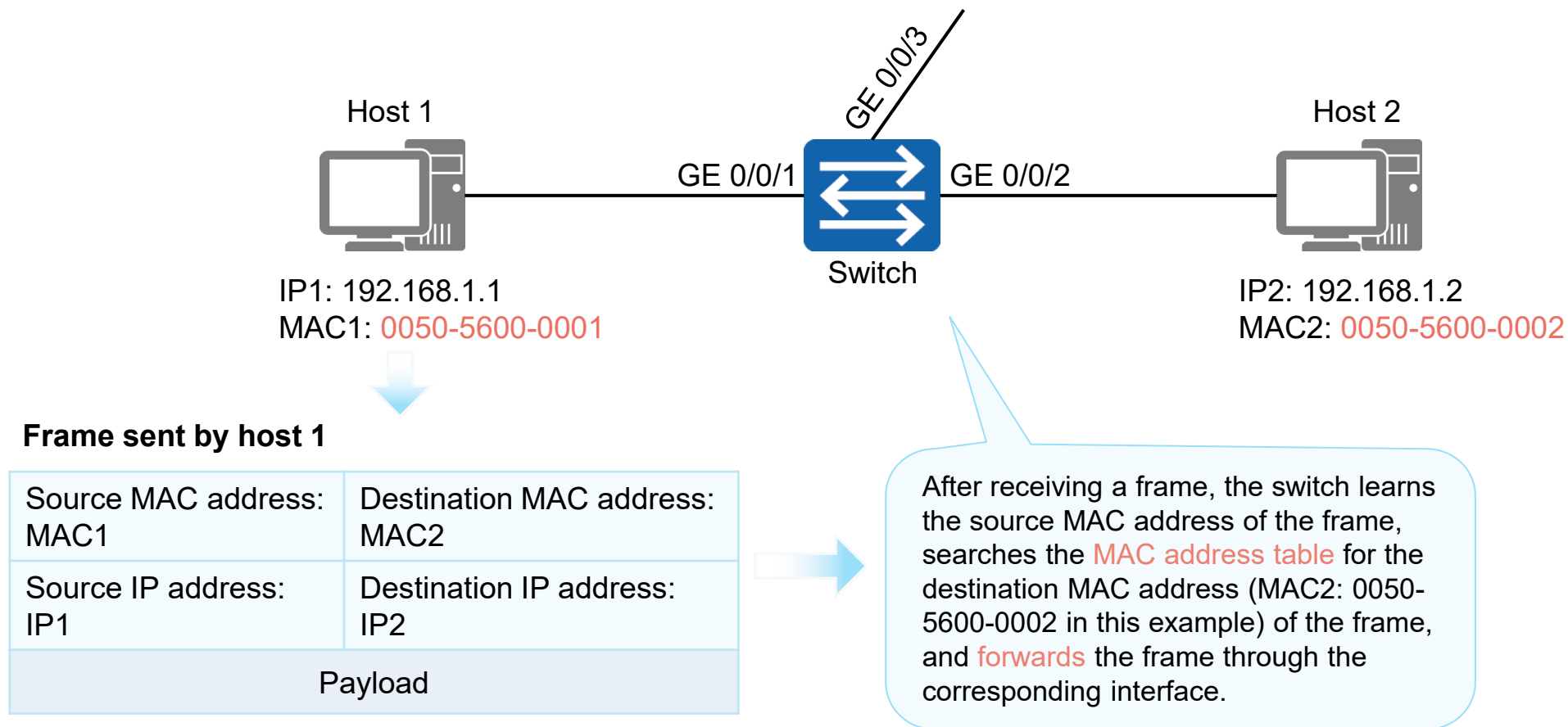
Layer 2 Ethernet switch

Layer 2 Ethernet switches forward data through Ethernet interfaces and can address and forward data only according to the MAC address in a Layer 2 header (Ethernet frame header).





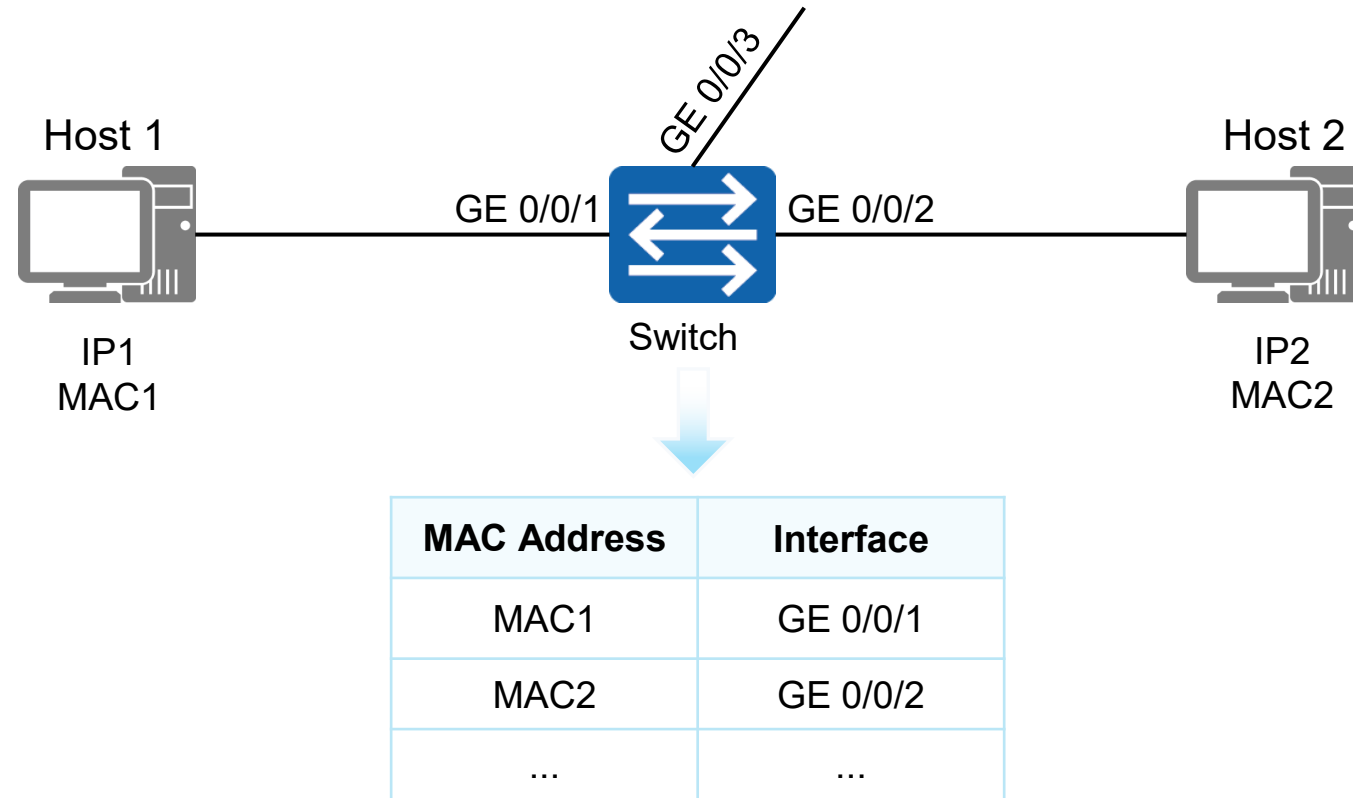
Working Principles of Switches





MAC Address Table

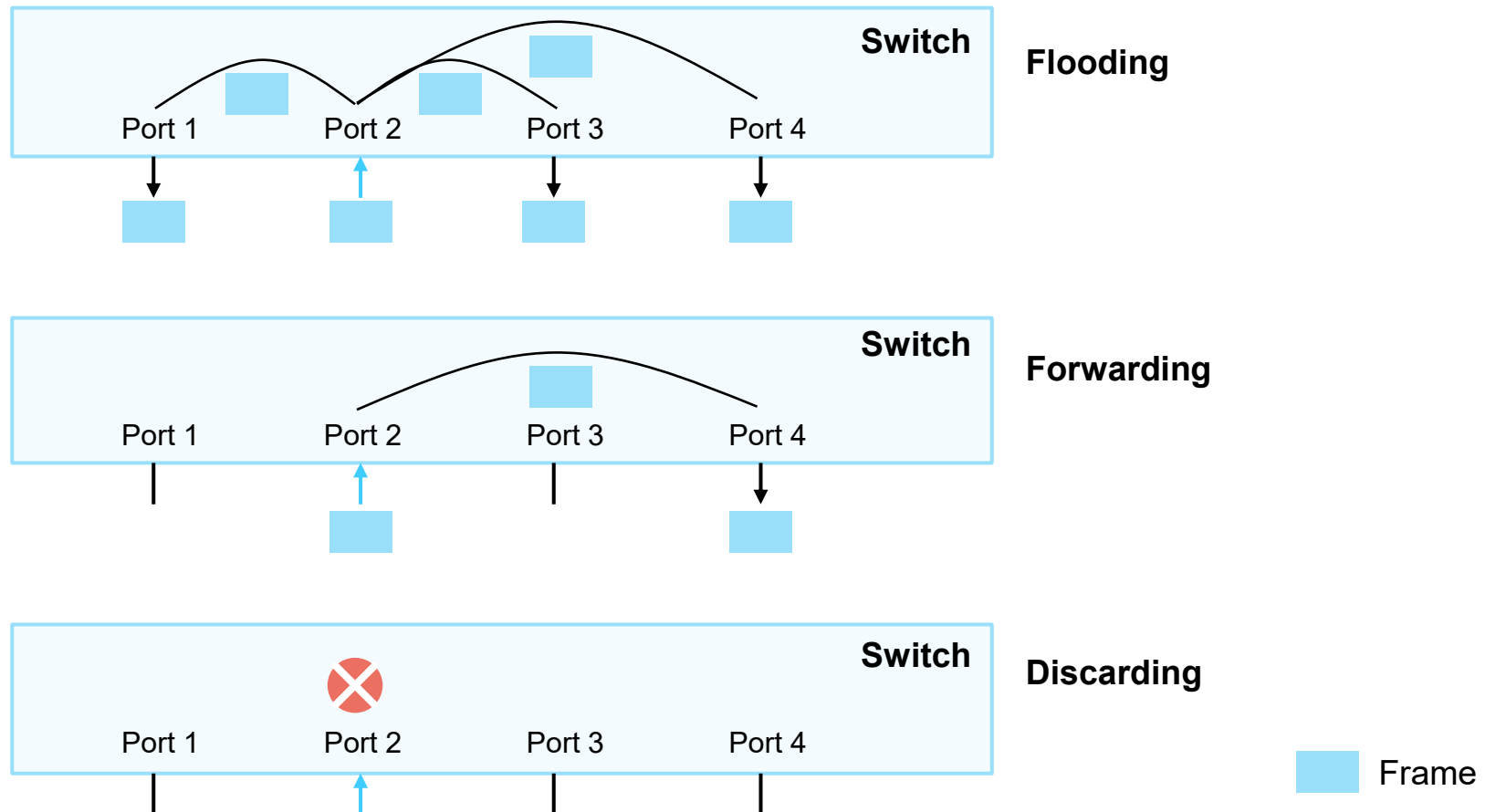
- Each switch has a MAC address table that stores the mapping between MAC addresses and switch interfaces.





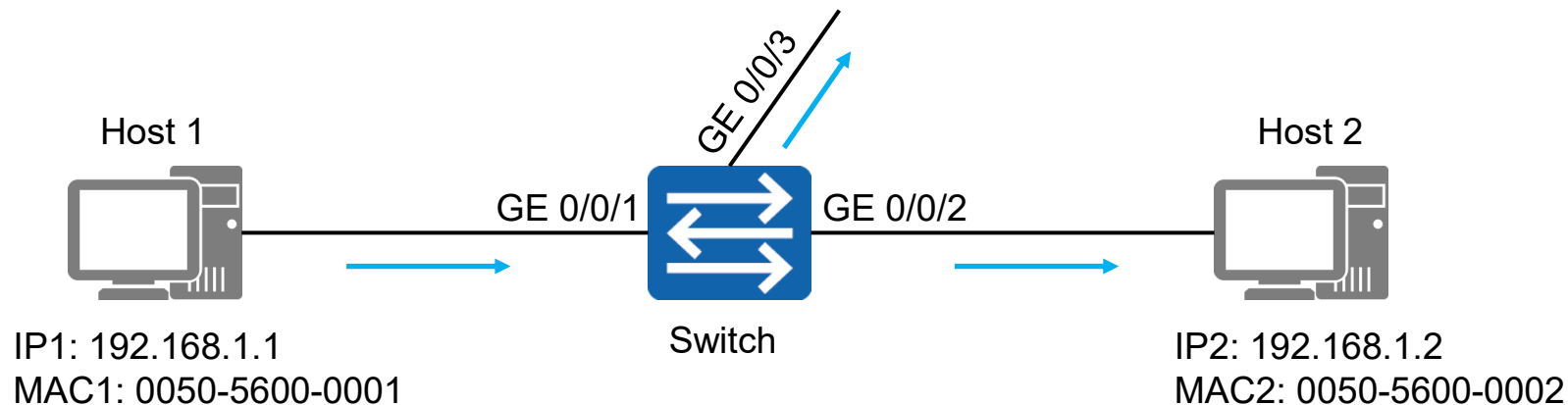
Three Frame Processing Behaviors of a Switch

- A switch processes the frames entering an interface over a transmission medium in three ways:





Flooding



1 Frame sent by host 1

Source MAC: MAC1
Destination MAC: **MAC2**

or

Source MAC: MAC1
Destination MAC: **FF-FF-FF-FF-FF-FF**

2 MAC address table searched by the switch

MAC Address	Interface
MAC1	GE 0/0/1

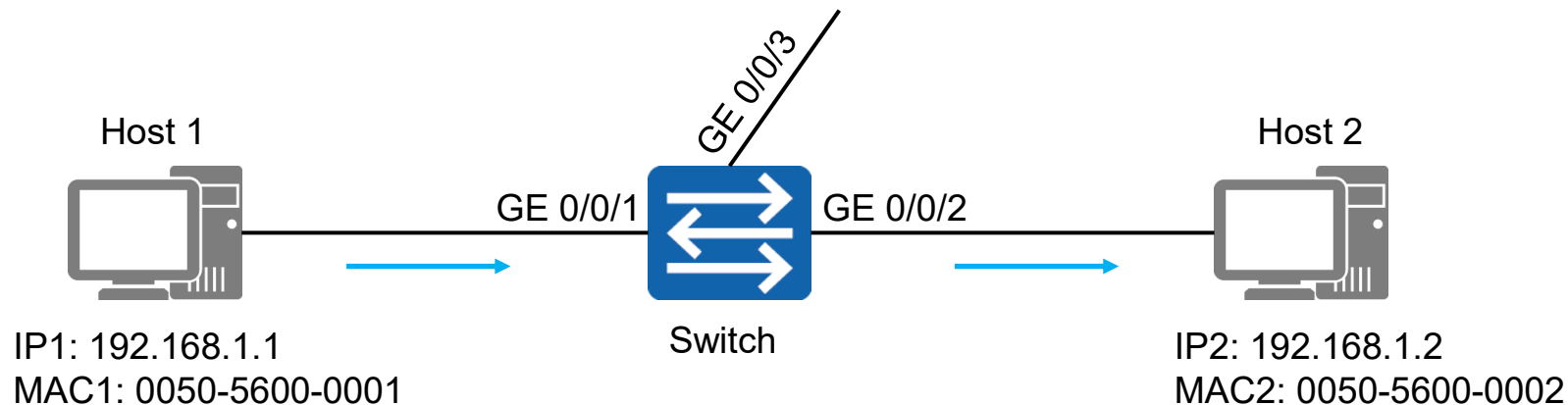
3 Frame processing behavior of the switch

- **If a unicast frame is received:**
If the switch cannot find the destination MAC address of the frame in the MAC address table, the switch floods the unicast frame.
- **If a broadcast frame is received:**
The switch directly floods the broadcast frame without searching the MAC address table.

Unknown unicast frame



Forwarding



1 Frame sent by host 1

Source MAC: MAC1
Destination MAC: MAC2

2 MAC address table searched by the switch

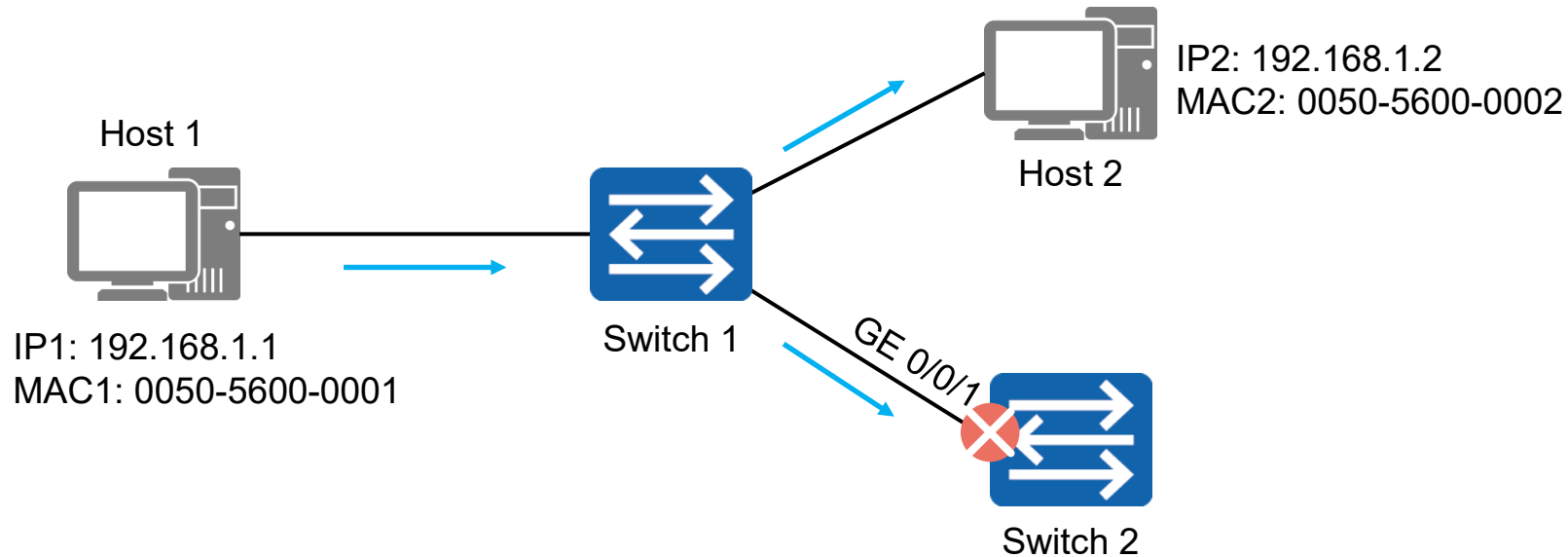
MAC Address	Interface
MAC1	GE 0/0/1
MAC2	GE 0/0/2

3 Frame processing behavior of the switch

- If a unicast frame is received:**
If the switch finds the destination MAC address of the frame in the MAC address table and the interface number in the table is not the number of the interface through which the frame enters over the transmission medium, the switch forwards the unicast frame.



Discarding



1 Frame sent by host 1

Source MAC: MAC1
Destination MAC: MAC2

2 MAC address table queried by switch 2

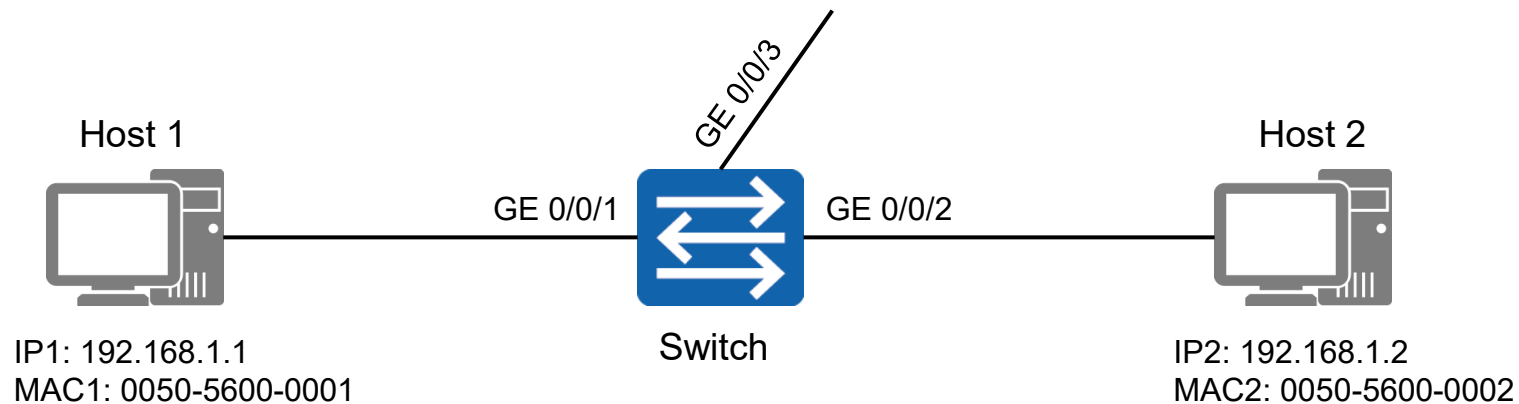
MAC Address	Interface
MAC2	GE 0/0/1

3 Frame processing behavior of the switch

- **If a unicast frame is received:**
- The switch finds the destination MAC address of the frame in the MAC address table, but the interface number in the table is the number of the interface through which the frame enters the switch over the transmission medium. In this case, the switch discards the unicast frame.



MAC Address Learning on a Switch (1)



MAC address table
of the switch

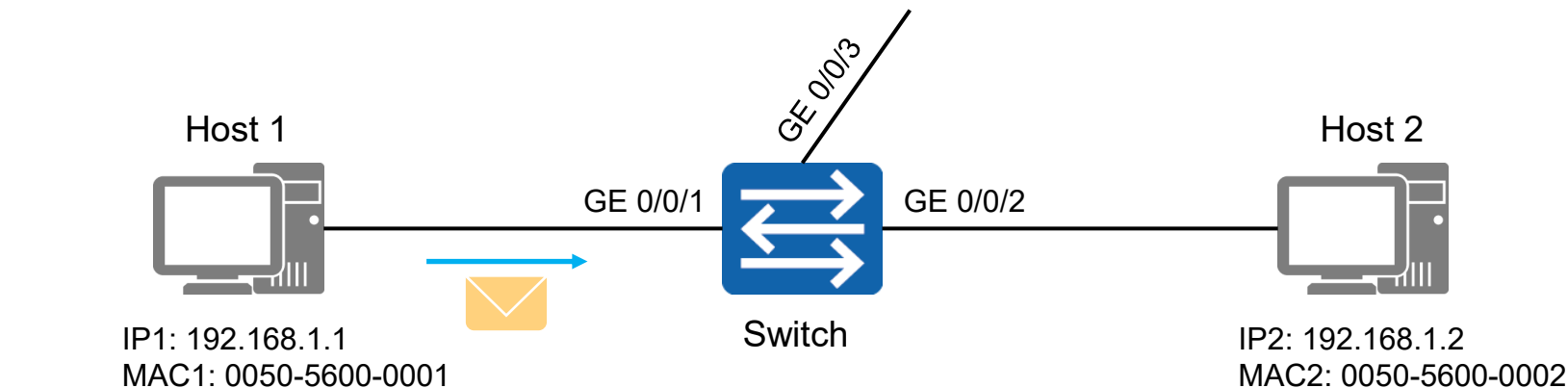
MAC Address	Interface

1

Initially, the MAC address table of the switch is empty.



MAC Address Learning on a Switch (2)



Frame sent by host 1

Source MAC: MAC1
Destination MAC: **MAC2**

(Assume that host 1 has obtained the MAC address of host 2.)

MAC address table searched by the switch

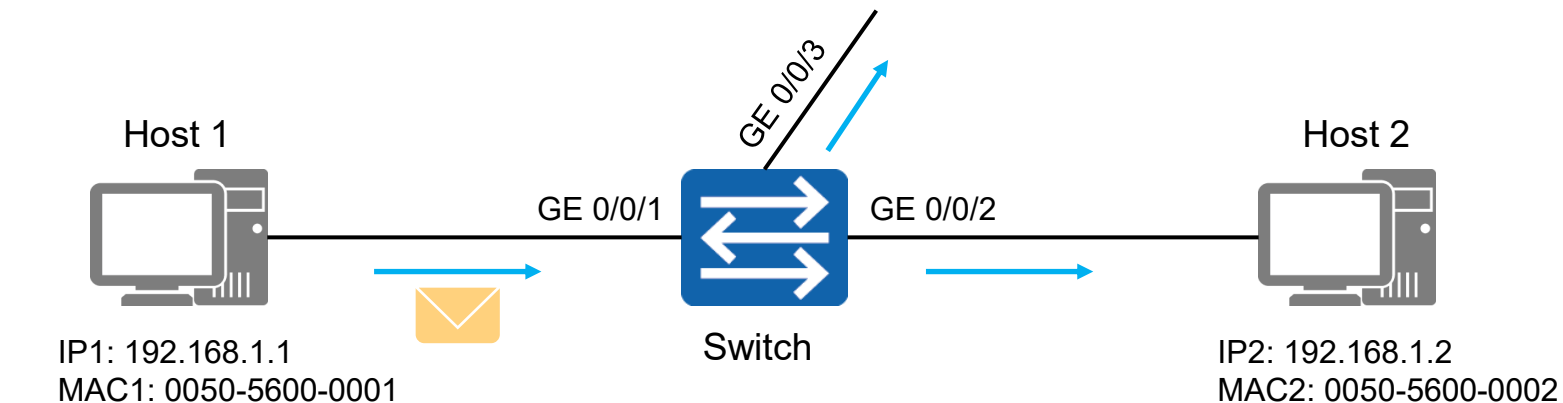
MAC Address	Interface

2

- Host 1 sends a frame to host 2.
- After the frame is received on the switch's GE 0/0/1, the switch searches the MAC address table for the destination MAC address of the frame. If no matching entry is found, the switch considers the frame an unknown unicast frame.



MAC Address Learning on a Switch (3)



Frame sent by host 1

Source MAC: **MAC1**
Destination MAC: MAC2

MAC address table searched by the switch

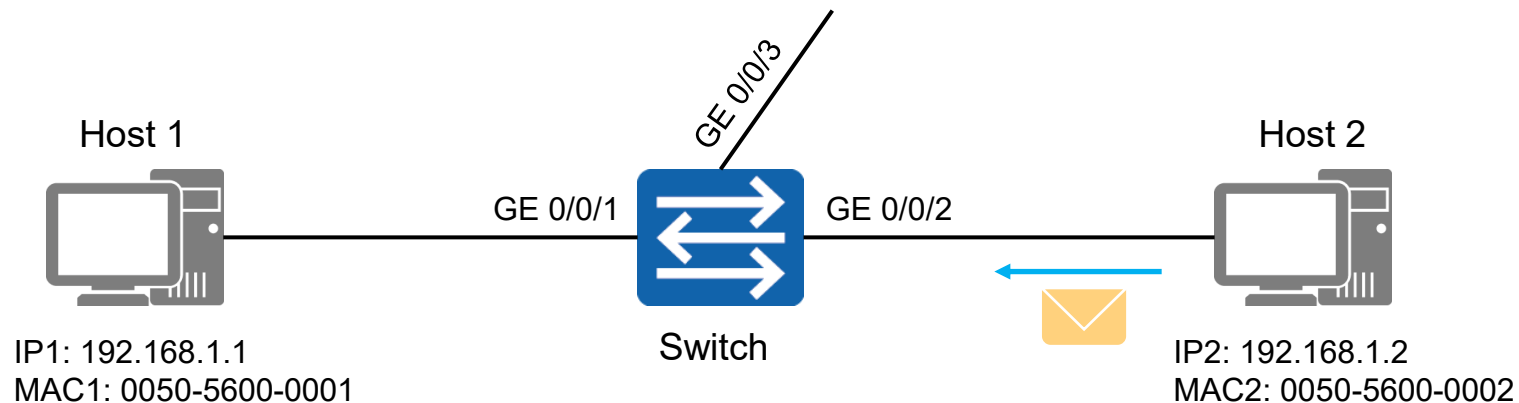
MAC Address	Interface
MAC1	GE 0/0/1

3

- If the corresponding entry is not found in the MAC address table, the switch **floods** the unicast frame.
- At the same time, the switch learns the source **MAC** address of the frame, creates the corresponding MAC address entry, and associates the MAC address entry with **GE 0/0/1**.



MAC Address Learning on a Switch (4)



4

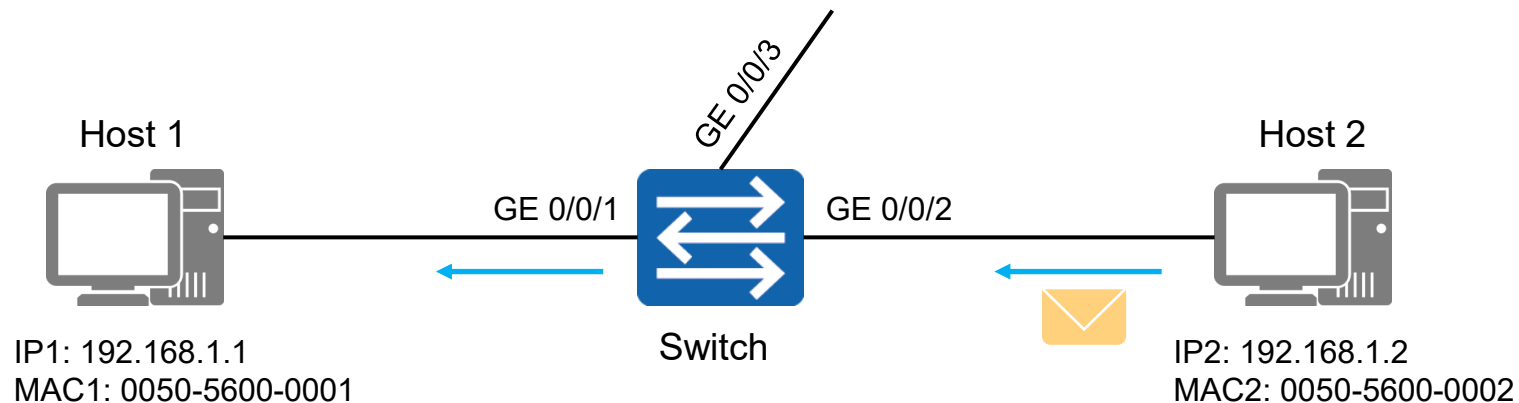
- The frame is also received by the hosts connected to other interfaces on the switch. These hosts, however, discard the frame.
- Host 2 receives and processes the frame, responds to host 1, and sends the frame to the switch.

Frame sent by host 2

Source MAC: MAC2
Destination MAC: MAC1



MAC Address Learning on a Switch (5)



MAC address table
searched by the switch

MAC Address	Interface
MAC1	GE 0/0/1
MAC2	GE 0/0/2

Frame sent by host 2

Source MAC: MAC2
Destination MAC: MAC1

5

- If the switch finds the corresponding entry in the MAC address table, the switch forwards the unicast frame through GE 0/0/1.
- At the same time, the switch learns the source MAC address of the frame, creates the corresponding MAC address entry, and associates the MAC address entry with GE 0/0/2.



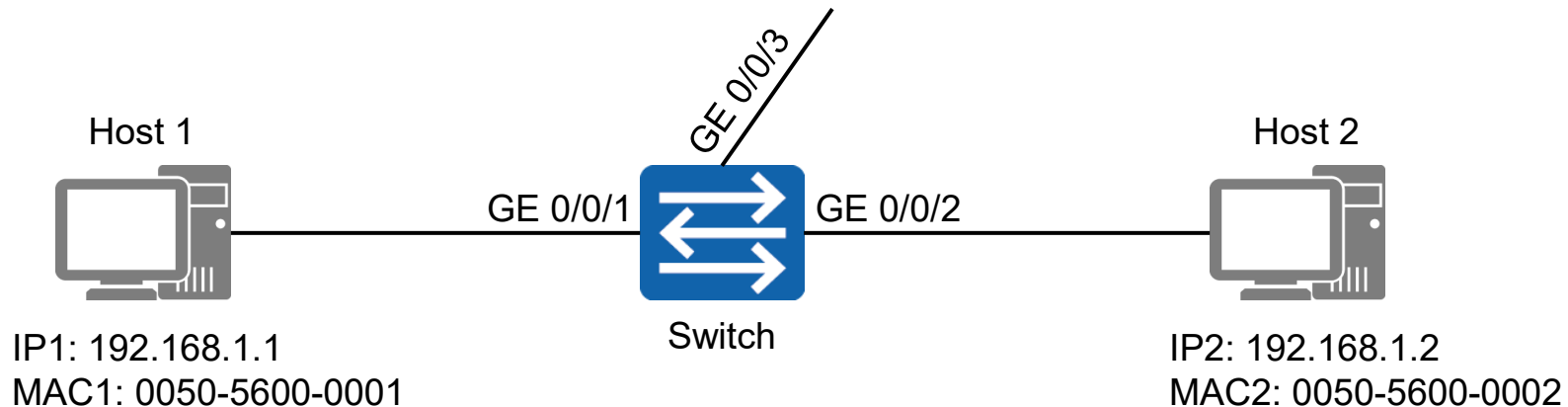
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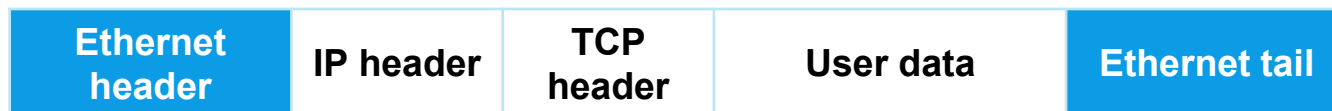
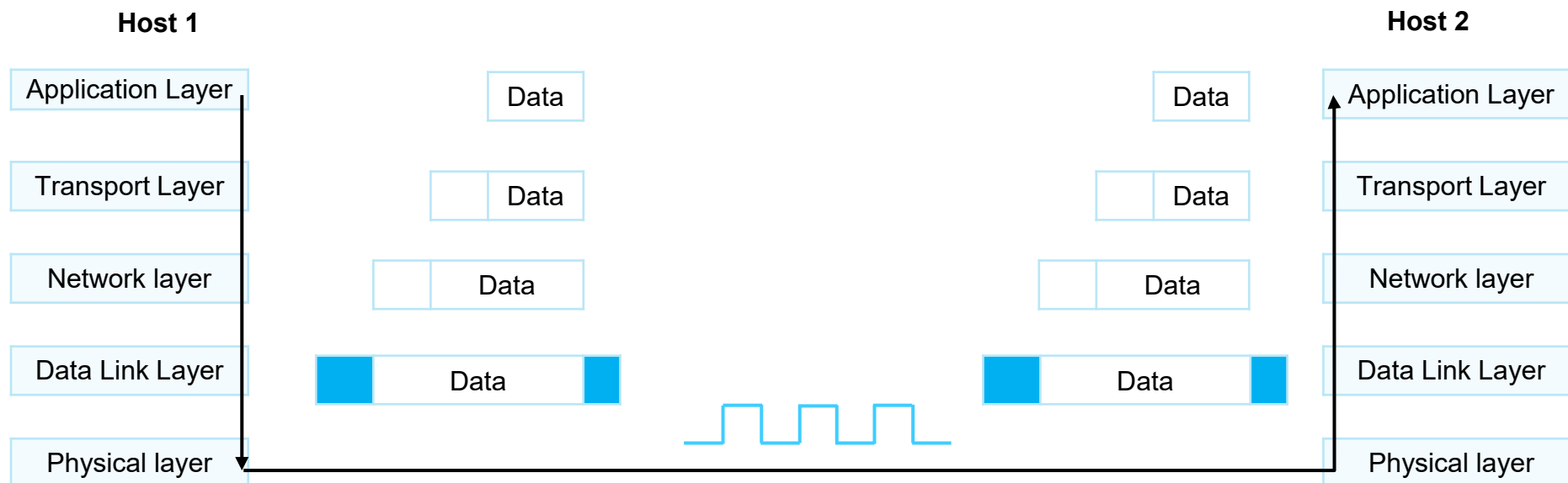
Process of Data Communication Within a Network Segment

- Scenario description:
 - ▢ Task: Host 1 wants to access host 2.
 - ▢ Host: The host is in the initialized state and only knows its own IP address and MAC address (assume that the IP address of the peer host has been obtained).
 - ▢ Switch: The switch is just powered on and in the initialized state.





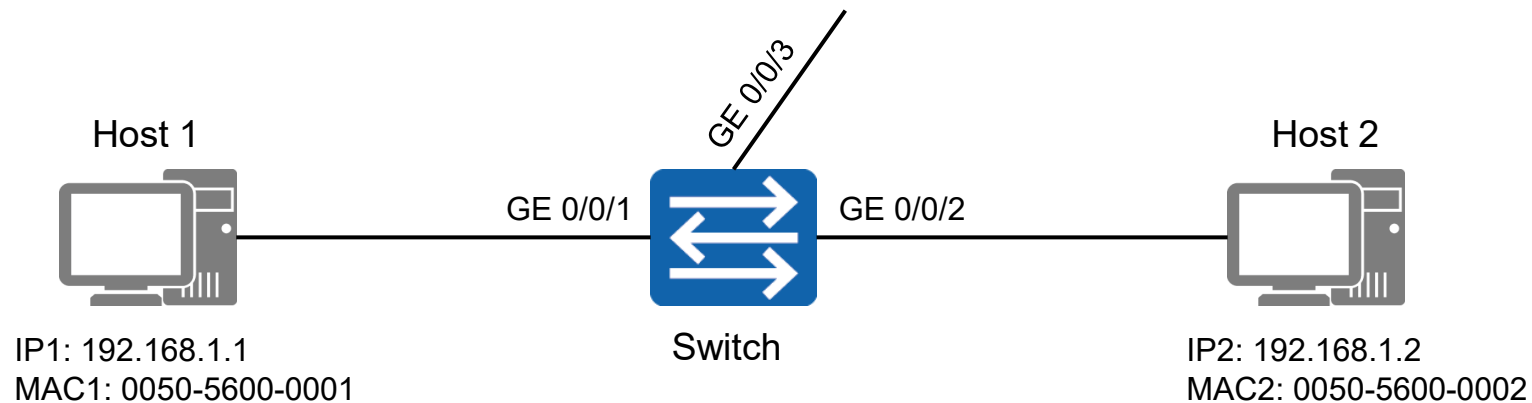
Data Encapsulation Process



- **Information that needs to be encapsulated:**
- Source MAC address
- Destination MAC address



Initialization



ARP cache table of host 1

```
Host 1>arp -a
Internet Address  Physical Address  Type
```

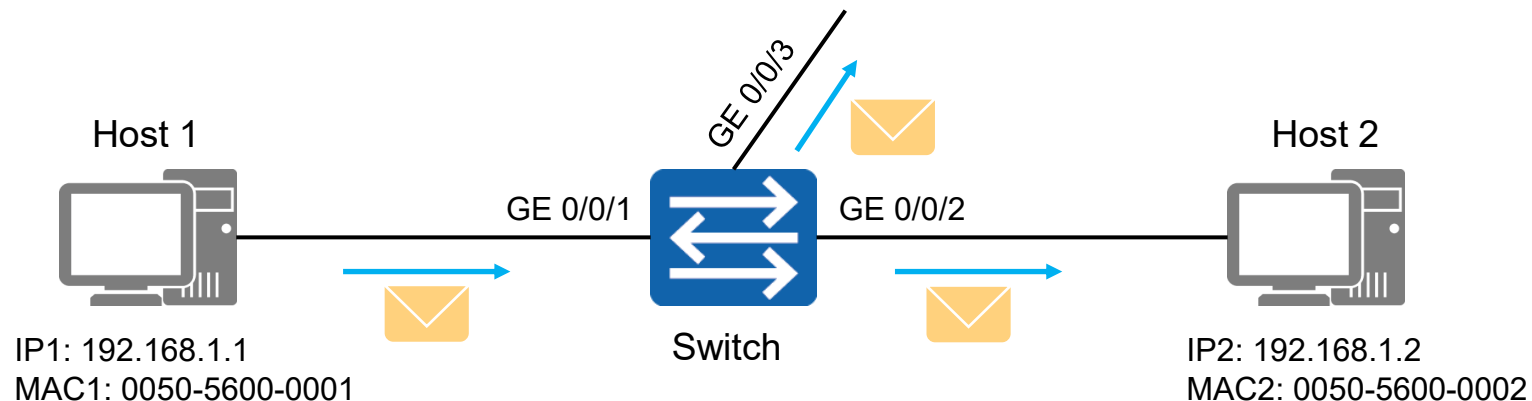
MAC address table of the switch

```
[Switch]display mac-address verbose
MAC address table of slot 0:
```

```
-----MAC
Address          Port          Type
```



Flooding Frames



ARP Request packet sent by host 1

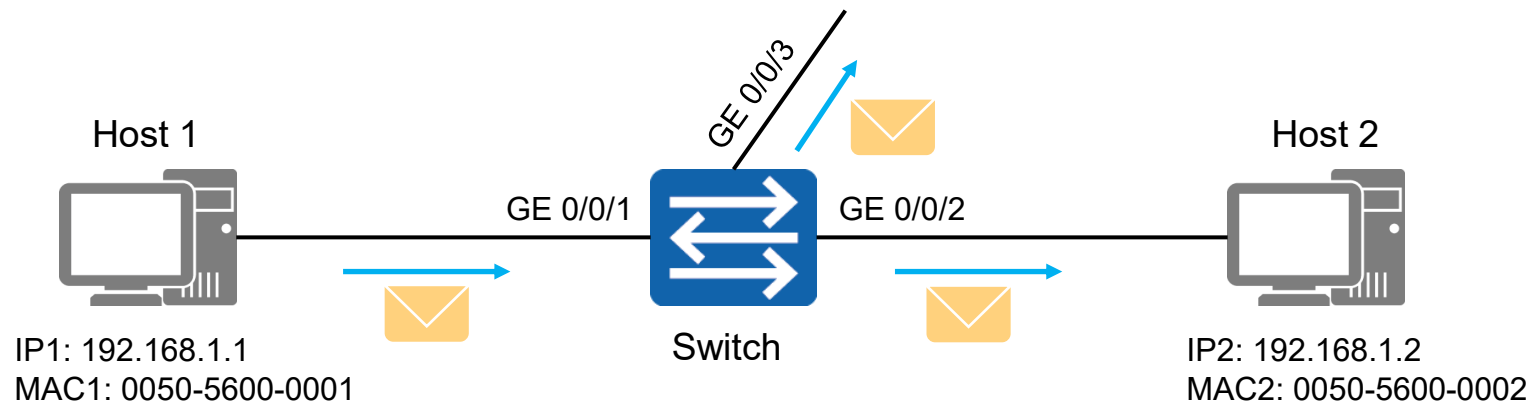
Source MAC address: MAC1	Destination MAC address: FF-FF-FF-FF-FF-FF
Source IP address: IP1	Destination IP address: IP2
Operation type: ARP Request Sender's MAC address: MAC1 Sender's IP address: IP1 Destination MAC address: 00-00-00-00-00-00 Destination IP address: IP2	

MAC address table of the switch

```
[Switch]display mac-address verbose
MAC address table of slot 0:
-----MAC
Address          Port      Type
-----
-----
```



MAC Address Learning



ARP Request packet sent by host 1

Source MAC address: MAC1	Destination MAC address: FF-FF-FF-FF-FF-FF
Source IP address: IP1	Destination IP address: IP2
Operation type: ARP Request Sender's MAC address: MAC1 Sender's IP address: IP1 Destination MAC address: 00-00-00-00-00-00 Destination IP address: IP2	

MAC address table of the switch

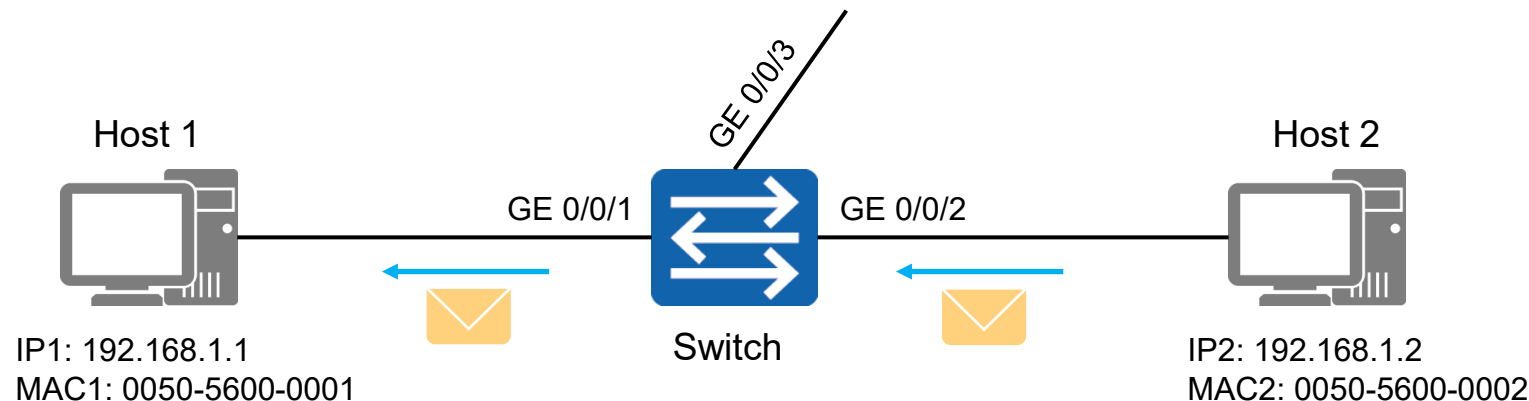
```
[Switch]display mac-address verbose
```

MAC address table of slot 0:

-----MAC		
Address	Port	Type
-----	-----	-----
5600-0001	GE0/0/1	dynamic
-----	-----	-----



Reply of the Target Host



MAC address table of the switch

```
[Switch]display mac-address verbose
MAC address table of slot 0:
-----MAC
Address          Port      Type
-----0050-
5600-0001        GE0/0/1   dynamic
0050-5600-0002   GE0/0/2   dynamic
-----
```

ARP Reply packet sent by host 2

Source MAC address: MAC2	Destination MAC address: MAC1
Source IP address: IP2	Destination IP address: IP1

Operation type: ARP Reply
Sender's MAC address: MAC2
Sender's IP address: IP2
Destination MAC address: MAC1
Destination IP address: IP1



Quiz

1. (Single) A Layer 2 Ethernet switch generates a MAC address table entry based on the () of the packet received by an interface.
 - A. Source MAC address
 - B. Destination MAC address
 - C. Source IP address
 - D. Destination IP address.
2. (Single) A switch has eight interfaces. A unicast frame enters the switch through one of the eight interfaces, but the switch cannot find the destination MAC address entry of the frame in the MAC address table. In this case, which of the following operations is performed by the switch? ()
 - A. Discarding
 - B. Flooding
 - C. Point-to-point forwarding



Summary

- This section describes the basic information about the Ethernet protocol, Ethernet frame formats, MAC address, and working principles of Layer 2 switches. Specifically, after receiving a frame, a switch learns the source MAC address of the frame and searches the destination MAC address of the frame in the MAC address table. If the destination MAC address exists in the table, the switch forwards the frame through the corresponding interface.
- This course also describes the whole process of data communication within the same network segment based on the working principles of switches.

The image features a blue-tinted background with silhouettes of several groups of business professionals. On the left, a group of four people is gathered around a table, looking at documents. In the center, two groups are visible: one of three people standing and talking, and another of four people, with one person holding a large folder or tablet. On the right, a group of four people is engaged in conversation. The entire scene is set against a backdrop of modern skyscrapers, creating a professional and corporate atmosphere.

Thank You
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